Vectors and Planes

Part 2/2: Angles, Distances and Intersections

CSC 1105 Mathematics for Computer Science

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Lecture Outline and Learning Outcomes

Lecture outline

- Angles involving lines and planes
- Parallelism and perpendicularity
- Distances from points to lines and planes
- Intersections and geometric relationships
- Applications in 3D graphics, animation and movies

Learning outcomes

1. Calculate angles and test geometric orientations.
2. Calculate shortest distances to lines and planes.
3. Find intersections and classify their outcomes.
4. Apply vector calculations to animation, collision checks and stereo views.
5. Implement a ray–plane intersection in Python.

Angle Between Two Lines

For nonzero direction vectors $\mathbf{d}_1$ and $\mathbf{d}_2$, the smaller angle is

$$\boxed{\cos\alpha = \frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{\|\mathbf{d}_1\|\,\|\mathbf{d}_2\|}}, \qquad 0 \le \alpha \le 90^\circ.$$

Example

Let $\mathbf{d}_1 = (1,1,0)^T$ and $\mathbf{d}_2 = (1,0,1)^T$.

$$\mathbf{d}_1 \cdot \mathbf{d}_2 = 1, \qquad \|\mathbf{d}_1\| = \|\mathbf{d}_2\| = \sqrt{2}.$$

$$\cos\alpha = \frac{1}{2} \quad \Longrightarrow \quad \boxed{\alpha = 60^\circ}.$$

Parallel directions give $\alpha = 0^\circ$. Orthogonal directions give $\alpha = 90^\circ$, but the lines are perpendicular only if they intersect.

Angle Between a Line and a Plane

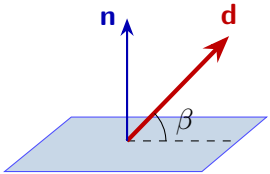
For nonzero line direction $\mathbf{d}$ and plane normal $\mathbf{n}$,

$$\boxed{\sin\beta = \frac{|\mathbf{d}\cdot\mathbf{n}|}{\|\mathbf{d}\|\,\|\mathbf{n}\|}}, \qquad 0 \le \beta \le 90^\circ.$$

Example: $\mathbf{d} = (1,0,1)^T$ and plane $z = 2$.

$$\mathbf{n} = (0,0,1)^T, \qquad \sin\beta = \frac{1}{\sqrt{2}}.$$

$$\boxed{\beta = 45^\circ}.$$



β complements the smaller angle between $\mathbf{d}$ and $\mathbf{n}$.

- $\beta = 0^\circ$: the line is parallel to or contained in the plane.
- $\beta = 90^\circ$: the line is perpendicular to the plane.

Angle Between Two Planes

For nonzero normals $\mathbf{n}_1$ and $\mathbf{n}_2$, the smaller angle is

$$\boxed{\cos\phi = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\|\,\|\mathbf{n}_2\|}}, \qquad 0 \le \phi \le 90^\circ.$$

Example

The planes $\Pi_1 : x + z = 2$ and $\Pi_2 : y + z = 3$ have normals

$$\mathbf{n}_1 = (1,0,1)^T, \qquad \mathbf{n}_2 = (0,1,1)^T.$$

$$\cos\phi = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2} \quad \Longrightarrow \quad \boxed{\phi = 60^\circ}.$$

- Parallel normals give parallel or coincident planes.
- Orthogonal normals give perpendicular planes.

Distance from a Point to a Plane

Let $P = (x_0, y_0, z_0)$ have position vector $\mathbf{p}$, and let

$$\Pi : ax + by + cz = k, \qquad \mathbf{n} = (a, b, c)^T \neq \mathbf{0}.$$

The shortest distance is measured along the normal:

$$\boxed{d(P, \Pi) = \frac{|\mathbf{n} \cdot \mathbf{p} - k|}{\|\mathbf{n}\|} = \frac{|ax_0 + by_0 + cz_0 - k|}{\sqrt{a^2 + b^2 + c^2}}}.$$

Example

For $P = (2, 3, 4)$ and $\Pi : x + y + z = 3$,

$$d = \frac{|2 + 3 + 4 - 3|}{\sqrt{1 + 1 + 1}} = \frac{6}{\sqrt{3}} = \boxed{2\sqrt{3}}.$$

Distance from a Point to a Line

For P with position vector $\mathbf{p}$ and $L : \mathbf{r} = \mathbf{a} + t\mathbf{d}$, where $\mathbf{d} \neq \mathbf{0}$,

$$\boxed{d(P, L) = \frac{\|(\mathbf{p} - \mathbf{a}) \times \mathbf{d}\|}{\|\mathbf{d}\|}}.$$

Example

Let $P = (2, 3, 1)$ and $L : \mathbf{r} = (1, 1, 1)^T + t(1, 1, 0)^T$.

$$\mathbf{p} - \mathbf{a} = (1, 2, 0)^T, \qquad (\mathbf{p} - \mathbf{a}) \times \mathbf{d} = (0, 0, -1)^T.$$

$$\boxed{d(P, L) = \frac{1}{\sqrt{2}}}.$$

Intersection of a Line and a Plane

For $L : \mathbf{r} = \mathbf{a} + t\mathbf{d}$ and $\Pi : \mathbf{n} \cdot \mathbf{r} = k$, if $\mathbf{n} \cdot \mathbf{d} \neq 0$,

$$\boxed{t = \frac{k - \mathbf{n} \cdot \mathbf{a}}{\mathbf{n} \cdot \mathbf{d}}}, \qquad \mathbf{r} = \mathbf{a} + t\mathbf{d}.$$

Example

Let $L : \mathbf{r} = (1, 0, 0)^T + t(1, 1, 1)^T$ and $\Pi : x + y + z = 4$. Substitution gives

$$(1 + t) + t + t = 4 \quad \Longrightarrow \quad t = 1, \qquad \boxed{P = (2, 1, 1)}.$$

When $\mathbf{n} \cdot \mathbf{d} = 0$

- $\mathbf{n} \cdot \mathbf{a} = k$: the line lies in the plane.
- $\mathbf{n} \cdot \mathbf{a} \neq k$: the line and plane are parallel and disjoint.

Intersection of Two Planes

For $\Pi_i : \mathbf{n}_i \cdot \mathbf{r} = k_i$ with nonparallel normals, the intersection is a line with direction $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$.

Example

$$\Pi_1 : x + y + z = 3, \qquad \Pi_2 : x - y + z = 1.$$

Subtracting gives $y = 1$ and $x + z = 2$. A common point is $(0, 1, 2)$. Since $\mathbf{n}_1 \times \mathbf{n}_2 = (2, 0, -2)^T$, use direction $(1, 0, -1)^T$.

$$\boxed{\mathbf{r} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \qquad t \in \mathbb{R}.}$$

If $\mathbf{n}_2 = \lambda \mathbf{n}_1$, the planes coincide when $k_2 = \lambda k_1$. Otherwise, they are distinct parallel planes.

Intersection of Three Planes

Three planes give a linear system $A\mathbf{r} = \mathbf{b}$.

- $\det(A) \neq 0$: one common point.
- $\det(A) = 0$: row reduction distinguishes no solution from infinitely many.

Example

$$\begin{cases} x + y + z = 6, \\ x - y + z = 2, \\ x + y - z = 4. \end{cases}$$

Subtract the second and third equations from the first:

$$2y = 4 \Longrightarrow y = 2, \qquad 2z = 2 \Longrightarrow z = 1.$$

Then $x = 6 - 2 - 1 = 3$, so $\boxed{(x, y, z) = (3, 2, 1)}$.

Intersection of Two Lines in 3D

For $L_1 : \mathbf{r} = \mathbf{a} + t\mathbf{u}$ and $L_2 : \mathbf{r} = \mathbf{b} + s\mathbf{v}$, solve

$$\mathbf{a} + t\mathbf{u} = \mathbf{b} + s\mathbf{v}$$

using all three coordinate equations.

Example

Let $L_1 : \mathbf{r} = t(1,1,0)^T$ and $L_2 : \mathbf{r} = (2,0,0)^T + s(-1,1,0)^T$.

$$t = 2 - s, \qquad t = s, \qquad 0 = 0 \quad \Longrightarrow \quad t = s = 1.$$

The intersection is $\boxed{(1,1,0)}$.

- If $\mathbf{u} \times \mathbf{v} \neq \mathbf{0}$ and the system has no solution, the lines are **skew**.
- If $\mathbf{u} \times \mathbf{v} = \mathbf{0}$, the lines coincide when $(\mathbf{b} - \mathbf{a}) \times \mathbf{u} = \mathbf{0}$. Otherwise, they are distinct parallel lines.

Animation: Moving Between Two Keyframes

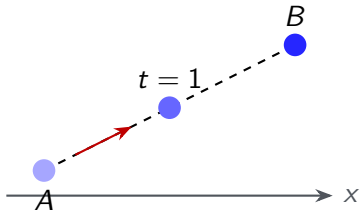
Move an object from A to B over $T > 0$ seconds. For $0 \le t \le T$, let $s = t/T$:

$$\boxed{\mathbf{p}(s) = (1-s)\mathbf{a} + s\mathbf{b}}.$$

For $A = (1, 0, 0)$, $B = (5, 2, 0)$ and $T = 2$ seconds, at $t = 1$ second we have $s = 0.5$:

$$\mathbf{p}(0.5) = 0.5\mathbf{a} + 0.5\mathbf{b} = \boxed{(3, 1, 0)^T}.$$

Linear timing gives constant speed along the straight path.



Positions in the xy-plane

Animation: Rotations and Joint Angles

A 90° turn about the z-axis

$$R_z(90^\circ) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

For vertex $P = (2, 0, 1)$,

$$\boxed{R_z(90^\circ) \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}}.$$

For rotation about a centre $\mathbf{c}$,

$$\mathbf{p}' = \mathbf{c} + R_z(\theta)(\mathbf{p} - \mathbf{c}).$$

A joint angle

Take vectors from a joint towards its neighbouring joints:

$$\mathbf{u} = (1, 0, 0)^T, \qquad \mathbf{v} = (1, 1, 0)^T.$$

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|} = \frac{1}{\sqrt{2}}.$$

$$\boxed{\theta = 45^\circ}.$$

Compare θ with the joint's allowed range when checking the pose.

Collision Checks: A Ball and the Floor

A ball of radius $R = 0.5$ has centre $P = (2, 1, 0.3)$ above the floor $z = 0$.
The centre's distance to the floor is

$$d = |0.3| = 0.3 < R.$$

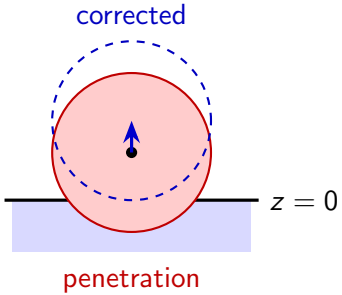
The ball penetrates the floor by

$$\boxed{R - d = 0.5 - 0.3 = 0.2}.$$

A positional correction moves the centre upward to

$$\boxed{P' = (2, 1, 0.5)}.$$

The correction removes overlap. A bounce also requires a velocity response.



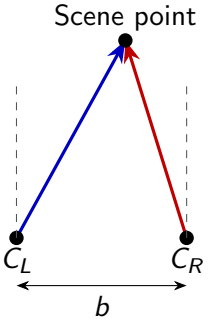
3D Movies: Stereo Viewpoints and Depth

Parallel pinhole cameras separated horizontally by baseline b render one view per eye. For identical focal length f, depth $Z > 0$ and centred image coordinates, the disparity is

$$\delta = u_L - u_R = \frac{fb}{Z}.$$

Here u_L and u_R are the horizontal coordinates of the same scene point. For $f = 1000$ pixels and $b = 0.20$ m:

Depth Z	**Disparity** δ
5 m	40 pixels
10 m	20 pixels



Reference: OpenCV: Camera Calibration and 3D Reconstruction. Ideal parallel pinhole camera model.

Python: A Camera Ray Meets the Floor

The ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$ has $t \geq 0$.

```
import numpy as np

o = np.array([1.0, 2.0, 5.0])
d = np.array([0.0, 0.0, -1.0])
n = np.array([0.0, 0.0, 1.0])
k = 0.0 # floor: n @ r = k
tol = 1e-9
denom = n @ d
if abs(denom) < tol:
    contained = abs(n @ o - k) < tol
    print("Ray lies in plane" if contained else "No hit")
else:
    t = (k - n @ o) / denom
    print(o + t*d if t >= 0 else "No forward hit")
```

For a triangle mesh, also check whether the hit lies inside the face.

Questions?